



Worked example in R



WorkedExample_GLMM_count.R

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Beall webworm data

- Counts of webworms in plots sprayed with insecticide or lead/arsenic
 - 2 insecticide treatments (N/Y)
 - 2 lead treatments (N/Y)
 - 2 x 2 factorial design



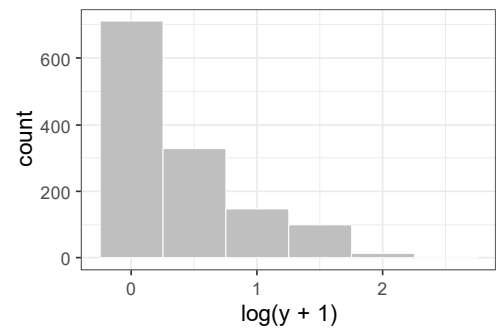
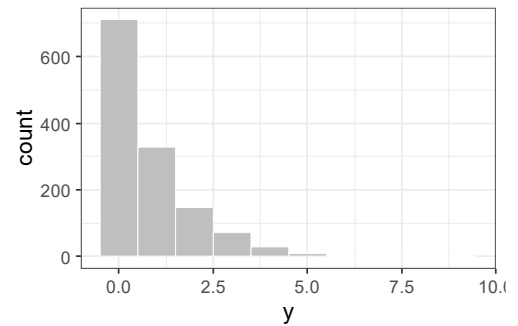
	row	col	y	block	trt	spray	lead
1	1	1	1	B1	T1	N	N
2	2	1	0	B1	T1	N	N
3	3	1	1	B1	T1	N	N
4	4	1	3	B1	T1	N	N
5	5	1	6	B1	T1	N	N
6	6	1	0	B2	T1	N	N
7	7	1	2	B2	T1	N	N
8	8	1	2	B2	T1	N	N
9	9	1	1	B2	T1	N	N
10	10	1	3	B2	T1	N	N
11	11	1	0	B3	T1	N	N
12	12	1	3	B3	T1	N	N
13	13	1	1	B3	T1	N	N
14	14	1	0	B3	T1	N	N
15	15	1	2	B3	T1	N	N
16	16	1	0	B4	T1	N	N

Showing 1 to 16 of 1,300 entries, 7 total columns

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Beall webworm data

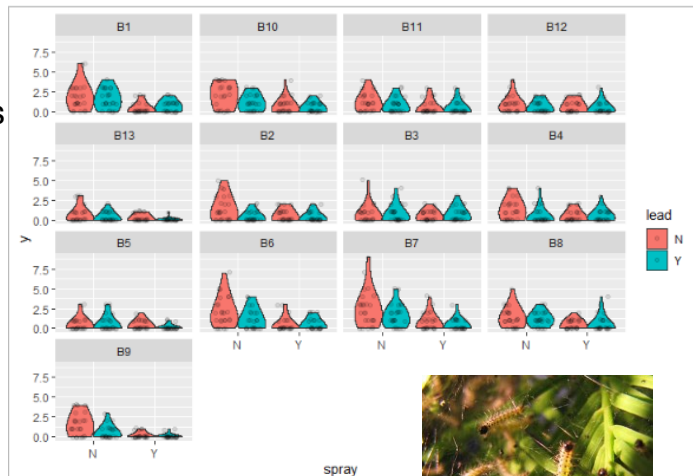
- Counts of webworms in plots sprayed with insecticide or lead/arsenic
 - 2 insecticide treatments (N/Y)
 - 2 lead treatments (N/Y)
 - 2 x 2 factorial design



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Beall webworm data

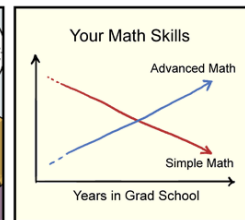
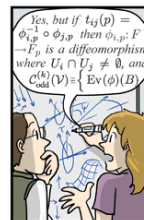
- Counts of webworms in plots sprayed with insecticide or lead/arsenic
 - 2 insecticide treatments (N/Y)
 - 2 lead treatments (N/Y)
 - 2 x 2 factorial design
- Plots setup in blocks
 - Many subplots per block



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Steps for constructing and interpreting a GLMM

0. Data Exploration
1. Construct model
 - a. Fixed effects (predictor or explanatory variables)
 - b. Random effects (blocking variables)
 - c. Error distribution (response variable or dependent variable)
2. Assess residuals and overdispersion
3. Check random effects and/or distribution (not always needed)
4. Examine significance of terms in model
 1. Anova
 2. AIC model selection
5. Calculate treatment means and/or slopes
6. Construct contrasts
7. Calculate R²m and R²c
8. Make pretty figure



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Step 1: Setting up a basic model in R

```
>
> library(glmmTMB)
> mod_logtrans <- glmmTMB(log(y+1) ~ spray * lead , data=d1)
```

$$\log(y_{ij}+1) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j) + \varepsilon_{ij}$$

* is shorthand for (spray + lead + spray:lead)

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Step 2: Examine residuals (check assumptions)

Conventional residuals are not always useful for GL(M)Ms

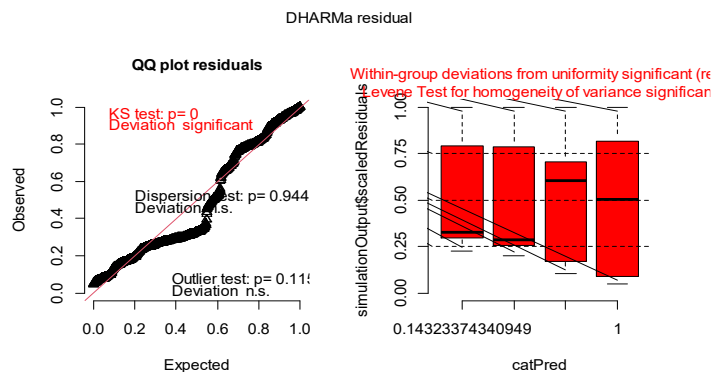
How to diagnose your fitted GL(M)M using DHARMA?

- The resulting residuals are standardized to values between 0 and 1
- Can be interpreted as intuitively as residuals from a linear regression.
 - **QQplot (points should line up against 1:1 line) – same as normal**
 - **Residual~predicted for homogeneity (residuals equal spread across predictor or among groups, no funneling, etc.) – same as normal**
 - Histogram (should be flat, not normally distributed) – Different than normal
 - Plus additional tests of residual structure

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Step 2: Examine residuals

```
> mod_logtrans <- glmmTMB(log(y+1) ~ spray * lead, data=d1) # log-transformed model, bad bad bad
> plot(simulateResiduals(mod_logtrans)) ## DHARMA package simulated residuals
```



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Step 1b. Setting up a GLM in R

```
>
> mod_poisson <- glmmTMB(y ~ spray * lead, family="poisson", data=d1)
> |
```

$$\log(\lambda_{ij}) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j)$$

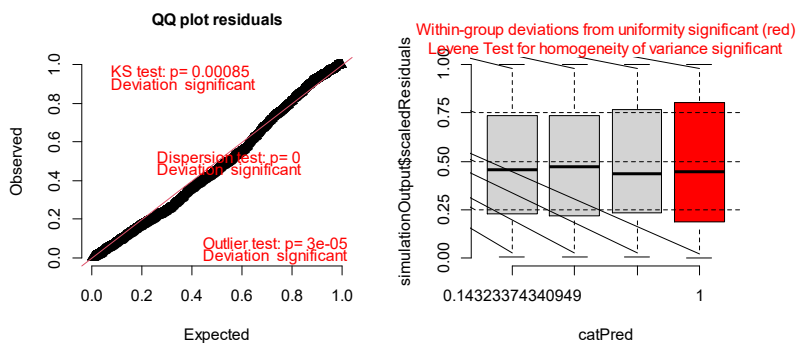
$$y_{ij} \sim \text{Poisson}(\lambda_{ij})$$

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Step 2b. Check residis and overdisperion

```
>
> mod_poisson <- glmmTMB(y ~ spray * lead, family="poisson", data=d1)
> |
```

DHARMA residual



```
> check_overdispersion(mod_poisson)
# overdispersion test
```

```
dispersion ratio = 1.355
Pearson's Chi-Squared = 1755.715
p-value = < 0.001
```

Overdispersion detected.

>1 indicates overdispersion,
Usually >2 is problematic

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What is overdispersion?

- More variation in the data than expected based on the distribution
- Poisson: $\lambda = \text{mean} = \text{variance}$
- Most count data: $\text{variance} > \text{mean}$
- Check: $\phi = \sum \text{Pearson Resid}^2 / \text{df}$
 - High values = more overdispersion
 - Alt: θ = overdispersion parameter
 - lower values are more overdispersion
 - higher values converge to Poisson
- Need to correct!

“Overdispersion is the polite statistician’s version of Murphy’s Law: if something can go wrong, it will”

Crawley (2007, p. 522)

Harrison 2014

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Processes that generate overdispersion

- Missing co-variates or random effects
- Sampling (counting) errors
- Non-independent (ie. **aggregated**) data
- Environmental stochasticity
- Many zeros (mixture of above, others, unknown)

Gelman and Hill 2007

Linden and Mantyniemi 2011 Ecology

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What are consequences of overdispersion?

- Confidence intervals too small
- P-values too low, leading to Type I error
- Biased parameter estimates

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How to remedy overdispersion?

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Step 1c. Update the GLM to correct overdisp.

```
> mod_nbinom <- glmmTMB(y ~ spray * lead, family="nbinom2", data=d1)
```

$$\log(\lambda_{ij}) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j)$$

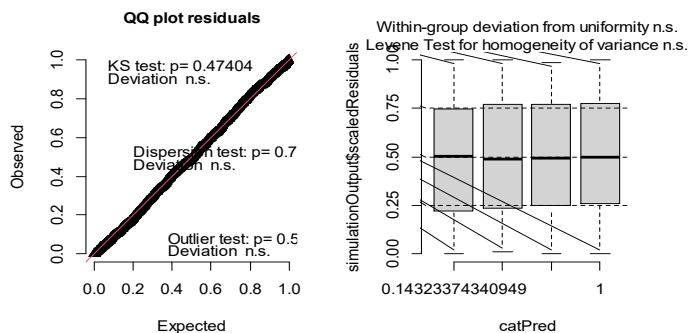
$$y_{ij} \sim \text{Negative Binomial}(\mu_{ij}, \phi_{ij})$$

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Step 2c. Check residis and overdispersion

```
> mod_nbinom <- glmmTMB(y ~ spray * lead, family="nbinom2", data=d1)
```

DHARMa residual



overdispersion test

dispersion ratio = 0.972
p-value = 0.752

No overdispersion detected.

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Looks good! What did we miss?

- Fixed effects
- Random effects
- Error distribution

Counts of webworms in plots sprayed with insecticide or lead/arsenic

2 insecticide treatments (N/Y)

2 lead treatments (N/Y)

2 x 2 factorial design

Plots setup in blocks

Many subplots per block

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Step 1d. Update the GLM to correct overdisp.

```
> mod_nbinom_blk <- glmmTMB(y ~ spray * lead + (1|block), data=d1, family="nbinom2")
```

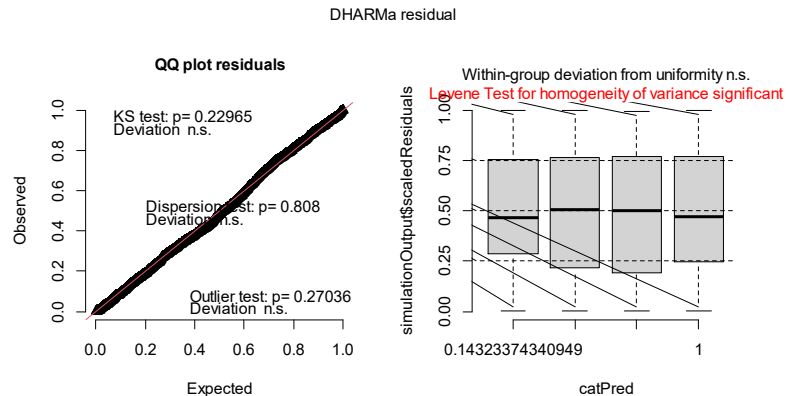
$$\log(\mu_{ijk}) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j) + b_k$$

$$y_{ij} \sim \text{Negative Binomial}(\mu_{ij}, \phi_{ij})$$

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2d. Check residuals

```
> mod_nbinom_blk <- glmmTMB(y ~ spray * lead + (1|block), data=d1, family="nbinom2")
> |
```



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The block is good... but is it the best?

- AIC is a measure of model fit, balancing explanatory power and simplicity
- Can use AIC to compare models with the same response variable
 - Models with different fixed and random effects
 - Can't compare:
 - Transformed vs untransformed
 - Models with different families
 - Full dataset vs subsetted dataset (ie. outlier removed)
- Lower AIC is better

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The block is good... but is it the best?

model	df	AIC
mod_nbinom	5	3052.969
mod_nbinom_blk	6	3007.412

df is the number of parameters estimated by the model

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3. Check random effects and model basics

```
> summary(mod_nbinom_blk)
Family: nbinom2 ( log )
Formula: y ~ spray * lead + (1 | block)
Data: d1

      AIC      BIC    logLik deviance df.resid 
3007.4    3038.4   -1497.7   2995.4     1294 

Random effects:
Conditional model:
Groups Name      variance Std.Dev.
block (Intercept) 0.107    0.3271
Number of obs: 1300, groups: block, 13

Dispersion parameter for nbinom2 family (): 2.8

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.28347    0.10813   2.622  0.00875 **
sprayY       -1.01163    0.10331  -9.792 < 2e-16 ***
leadY        -0.50041    0.09044  -5.533 3.14e-08 ***
sprayY:leadY  0.29143    0.15531   1.876  0.06060 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
```

Model basics

AIC, etc. (not all that helpful)

Random effect “variance component”

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3. Check random effects and model basics

```
> summary(mod_nbinom_blk)
Family: nbinom2 ( log )
Formula: y ~ spray * lead + (1 | block)
Data: d1
```

Model basics

```

      AIC      BIC    logLik deviance df.resid
3007.4    3038.4   -1497.7   2995.4     1294
```

AIC, etc. (not all that helpful)

Random effects:

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Groups Name      variance Std.Dev.
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Random effect “variance component”

Dispersion parameter for nbinom2 family (): 2.8

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Conditional model:
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(Intercept)  0.28347    0.10813   2.622  0.00875 **
sprayY       -1.01163    0.10331  -9.792 < 2e-16 ***
leadY        -0.50041    0.09044  -5.533 3.14e-08 ***
sprayY:leadY  0.29143    0.15531   1.876  0.06060 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

σ^2

```
> blk_vals
# A tibble: 13 x 5
  effect level term      estimate std.error
  <chr>   <chr>   <chr>         <dbl>    <dbl>
1 ran_vals B1    (Intercept)    0.333    0.142
2 ran_vals B10   (Intercept)    0.307    0.142
3 ran_vals B11   (Intercept)   -0.0405   0.150
4 ran_vals B12   (Intercept)   -0.129    0.154
5 ran_vals B13   (Intercept)   -0.580    0.177
6 ran_vals B2    (Intercept)   -0.0864   0.152
7 ran_vals B3    (Intercept)    0.00470   0.150
8 ran_vals B4    (Intercept)   -0.0423   0.150
9 ran_vals B5    (Intercept)   -0.458    0.169
10 ran_vals B6   (Intercept)    0.273    0.143
11 ran_vals B7   (Intercept)    0.504    0.139
12 ran_vals B8   (Intercept)    0.171    0.145
13 ran_vals B9   (Intercept)   -0.210    0.155
```

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3. Check random effects and model basics

```
> summary(mod_nbinom_blk)
Family: nbinom2 ( log )
Formula: y ~ spray * lead + (1 | block)
Data: d1
```

Model basics

```

      AIC      BIC    logLik deviance df.resid
3007.4    3038.4   -1497.7   2995.4     1294
```

AIC, etc. (not all that helpful)

Random effects:

```
Conditional model:
Groups Name      variance Std.Dev.
block (Intercept) 0.107    0.3271
Number of obs: 1300, groups: block, 13
```

Random effect “variance component”

Dispersion parameter for nbinom2 family (): 2.8

Dispersion parameter (θ)
Note: this is inverse of phi (ϕ):
 $\phi = 1/\theta$

```
Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.28347    0.10813   2.622  0.00875 **
sprayY       -1.01163    0.10331  -9.792 < 2e-16 ***
leadY        -0.50041    0.09044  -5.533 3.14e-08 ***
sprayY:leadY  0.29143    0.15531   1.876  0.06060 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Parameter estimates on log-scale
Not all that helpful in this format,
but used to calculate group means
(see Step 5)

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Step 4. Check significance

- Wald X2 test can be interpreted just like a regular ANOVA table
 - Anova() function is from car package; other packages will give different results

```
> # STEP 4. Check significance ####
> Anova(mod_nbinom_blk) # Anova function from car package
Analysis of Deviance Table (Type II wald chisquare tests)

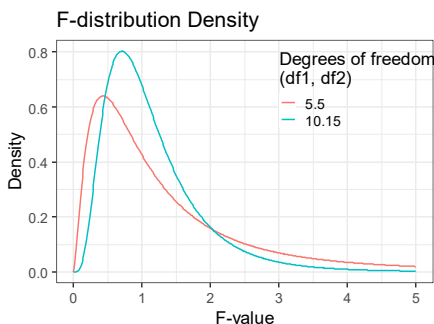
Response: y
          Chisq Df Pr(>Chisq)
spray      130.8698  1 < 2.2e-16 ***
lead       29.8260  1  4.726e-08 ***
spray:lead   3.5209  1   0.0606 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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Where do p-values come from?

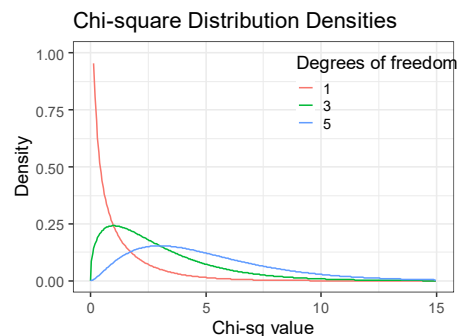
F-value

- From least squares
- $F = MS_B / MS_W$
- Calculate “p” from F-dist



Chi-sq value

- From maximum likelihood
- $\chi^2 = (\text{effect size} / \text{SE})^2$
- Calculate P from X2-dist



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A note about p-values

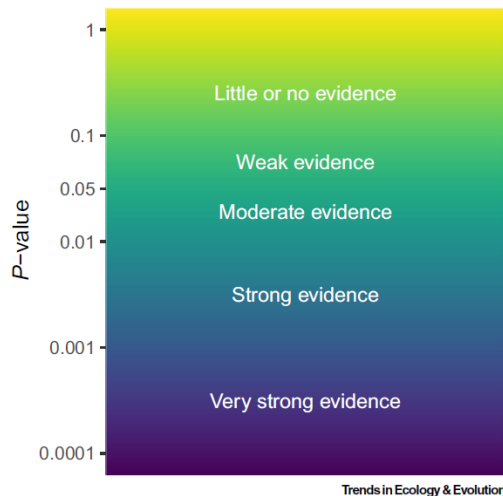
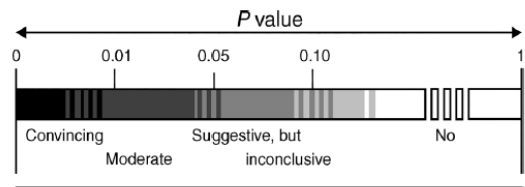


Figure 1. Suggested ranges to approximately translate the *P*-value into the language of evidence. The ranges are based on Bland (1986) [27], but the boundaries should not be understood as hard thresholds.



Is there evidence of a difference?

FIG. 1. Interpretation of the *P* value. Reprinted with permission from Ramsey and Schafer (2002).

Muff et al. 2022 <https://doi.org/10.1016/j.tree.2021.10.009>

Murtaugh 2014 <https://doi.org/10.1890/13-0590.1>

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Step 5. Calculate treatment means

```
> emmeans(mod_nbinom_blk, ~ spray:lead) # means on log-scale
  spray lead emmean    SE df asymp.LCL asymp.UCL
N      N    0.283 0.108 Inf    0.0715   0.49540
Y      N   -0.728 0.125 Inf   -0.9732  -0.48317
N      Y   -0.217 0.115 Inf   -0.4416   0.00775
Y      Y   -0.937 0.130 Inf   -1.1928  -0.68152
```

Results are given on the log (not the response) scale.
Confidence level used: 0.95

```
> |
```

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Step 5. Calculate treatment means (backtransformed)

```
> emmeans(mod_nbinom_blk, ~ spray:lead, type="response") # means on count-scale
  spray lead response      SE df asymp.LCL asymp.UCL
N      N      1.328 0.1436 Inf      1.074      1.641
Y      N      0.483 0.0603 Inf      0.378      0.617
N      Y      0.805 0.0923 Inf      0.643      1.008
Y      Y      0.392 0.0511 Inf      0.303      0.506

Confidence level used: 0.95
Intervals are back-transformed from the log scale
> |
```

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Step 6. Conduct contrasts

```
> emmeans(mod_nbinom_blk, pairwise ~ spray:lead, type="response") # means back-transformed
$emmeans
  spray lead response      SE df asymp.LCL asymp.UCL
N      N      1.328 0.1436 Inf      1.074      1.641
Y      N      0.483 0.0603 Inf      0.378      0.617
N      Y      0.805 0.0923 Inf      0.643      1.008
Y      Y      0.392 0.0511 Inf      0.303      0.506

Confidence level used: 0.95
Intervals are back-transformed from the log scale

$constrasts
  contrast ratio      SE df null z.ratio p.value
N N / Y N  2.75 0.284 Inf    1  9.792 <.0001
N N / N Y  1.65 0.149 Inf    1  5.533 <.0001
N N / Y Y  3.39 0.372 Inf    1 11.123 <.0001
Y N / N Y  0.60 0.066 Inf    1 -4.648 <.0001
Y N / Y Y  1.23 0.156 Inf    1  1.655 0.3480
N Y / Y Y  2.05 0.238 Inf    1  6.209 <.0001

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log scale
```

Note: in this case there are only 4 treatment combinations, so comparing them all is probably reasonable. In other cases, just make sure to think about your hypotheses and if it makes sense to compare them all, or if you only want to make a subset of comparisons.

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Step 6. Conduct contrasts - letters

```
> mod_emm <- emmeans(mod_nbinom_blk, ~ spray:lead, type="response")
> multcomp::cld(mod_emm, Letters="ABCD", alpha=.05)
```

spray	lead	response	SE	df	asympt.LCL	asympt.UCL	.group
Y	Y	0.392	0.0511	Inf	0.303	0.506	A
Y	N	0.483	0.0603	Inf	0.378	0.617	A
N	Y	0.805	0.0923	Inf	0.643	1.008	B
N	N	1.328	0.1436	Inf	1.074	1.641	C

```
Confidence level used: 0.95
Intervals are back-transformed from the log scale
P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log scale
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
> |
```

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Step 6. Conduct contrasts

- Lots more you can do with contrasts
 - <https://rvlenth.github.io/emmeans/articles/AQuickStart.html>
- Can also calculate slopes (when estimated) using emtrends from the emmeans package
 - <https://rvlenth.github.io/emmeans/reference/emtrends.html>

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7. Calculate R2m and R2c

- Marginal R2 (R^2_m): Variance explained by fixed effects
- Conditional R2 (R^2_c): Variance explained by fixed + random effects
- Intraclass correlation coefficient (ICC): Variance explained by random effects only
- $R2c \approx R2m + ICC$

```
> r2(mod_nbinom_b1k)
# R2 for Mixed Models

Conditional R2: 0.256
Marginal R2: 0.173
> icc(mod_nbinom_b1k)
# Intraclass Correlation Coefficient

Adjusted ICC: 0.100
Unadjusted ICC: 0.083
> |
```

[Nakagawa, S., Johnson, P. C. D., and Schielzeth, H. \(2017\). The coefficient of determination R2 and intra-class correlation coefficient from generalized linear mixed-effects models revisited and expanded. Journal of The Royal Society Interface, 14\(134\), 20170213.](#)

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Step 8. Make graphs

- Example plot here



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Steps for constructing and interpreting a GLMM

0. Data exploration
1. Construct model (glmmTMB works for everything)
 1. Fixed effects
 2. Random effects
 3. Error distribution
 4. Convergence okay?
2. Assess residuals and overdispersion
 1. Residual diagnostics with DHARMA::simulateResiduals
 2. Check overdispersion with check_overdispersion, correct using obs-level random effect or nbinom2
3. Check random effects and/or distribution using AIC or LRT (not always necessary)
4. Examine significance of terms in model (alt. AIC model selection for fixed effects)
 1. car::Anova
5. Calculate emmeans and/or emtrends
 1. Backtransform using type="response"
6. Construct contrasts (also emmeans/emmtrends)
7. Calculate R2m and R2c
 1. MuMin::r.squaredGLMM()
 2. Which one??? Log-normal ok w/ log link; delta always ok.
8. Make pretty figure

